Lesson Plan: $+c$?

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0.1 Introduction

An introductory course on differential calculus typically glosses over the mathematical origins of the constant of integration $+c$ which must be added onto the particular solution of an integral equation.

With a second-semester student of integral calculus in mind, I will explain the precise origins of the mysterious $+c$.

0.2 Exploration

0.2.1 Differential Equations

For some $n \in \mathbb{N}$, consider the following n^{th}-order, linear nonhomogeneous differential equation:

$$\sum_{k=0}^{n} q_k(t) y^{(k)}(t) = g(t), \qquad \forall t \in \Omega, \tag{1}$$

for known continuous functions $g, q_k : \Omega \to \mathbb{R}$ and unknown continuously n-differentiable function $y : \Omega \to \mathbb{R}$, all defined on some connected subspace $\Omega \subset R$.

We can assume $q_n(t) = 1$ because we can divide $g(t)$ and $p_k(t)$ by it on domains where $q_n(t) \neq 0$.

We associate to it the following homogenous differential equation:

$$\sum_{k=0}^{n} q_k(t) y_H^{(k)}(t) = 0, \qquad \forall t \in \Omega, \tag{2}$$

Suppose $\{y_k(t)\}_{k=1}^{n}$ is a fundamental set of solutions to Equation 2. Suppose $Y_1(t)$ and $Y_2(t)$ satisfy Equation 1, and let $Y(t) = Y_1(t) - Y_2(t)$.

Theorem 1. *There exist constants $c_k \in \mathbb{R}$ such that $Y(t) = \sum_{k=1}^{n} c_k y_k(t)$.*

Proof (Hernandez). For all $t \in \Omega$, we obtain the following equation:

$$\begin{aligned}
\sum_{k=0}^{n} q_k(t) Y^{(k)}(t) &= \sum_{k=0}^{n} q_k(t)(Y_1(t) - Y_2(t))^{(k)} && \text{by substitution} \\
&= \sum_{k=0}^{n} q_k(t)(Y_1^{(k)}(t) - Y_2^{(k)}(t)) && \text{by linearity} \\
&= \sum_{k=0}^{n} q_k(t) Y_1^{(k)}(t) - \sum_{k=0}^{n} q_k(t) Y_2^{(k)}(t) && \text{by associativity} \\
&= g(t) - g(t) = 0 && \text{by Equation 1.}
\end{aligned}$$

Thus, $Y(t)$ solves Equation 2. Our result follows from the fact that the $y_k(t)$ are a fundamental set of solutions to Equation 2. □

Let $y(t)$ be the general solution to Equation 1, and suppose $Y_P(t)$ is a known particular solution. Then, Theorem 1 shows $y(t) - Y_P(t) = y_c(t)$, where $y_c(t) = \sum_{k=1}^{n} c_k y_k(t)$ for some $c_k \in \mathbb{R}$, and thus that $y(t) = y_c(t) + Y_P(t)$.

In this way, solving the general equation $y(t)$ amounts to solving the homogeneous equation and finding some particular solution, both of which tend to be tractable problems.

0.2.2 Integral Calculus

Consider a first-order, linear nonhomogenous differential equation and its associated homogenous differential equation:

$$y'(t) = g(t), \qquad y'_H(t) = 0, \qquad \forall t \in \Omega, \tag{3}$$

as in Equation 1 with $n = 1$, $q_0(t) = 0$, and $q_1(t) = 1$. Given $g(t)$, we seek a function $y(t)$ which satisfies the following equivalent equations:

$$\int \frac{dy}{dt}\, dt = \int g(t)\, dt \qquad \text{by integrating Equation 3 with respect to} t \tag{4}$$

$$y(t) = \int g(t)\, dt \qquad \text{by evaluation.} \tag{5}$$

As in the previous section, $y(t) = Y_P(t)+y_c(t)$ where $Y_P(t)$ can be computed symbolically through various integration techniques and where $y_c(t) = c_1 y_1(t)$ is an arbitrary multiple of a solution $y_1(t)$ to the homogeneous equation. Notice $y'_H(t) = 0$ can be satisfied by $y_1(t) = 1$ or any other constant function.

Thus, $y(t) = Y_P(t) + c$. This is precisely where the constant $+c$ of integration comes from.

0.3 Summary

It is easy to show calc 1 students a reversed argument for why integral solutions are invariant under the addition of a constant. Here, I have shown calc 2 students where this constant comes from.

We have chosen to present this material so that the audience will feel familiar with such techniques as it is common in later courses on linear algebra and partial differential equations.